

Empowering Digital Agriculture: A Privacy-Preserving Framework for Data Sharing and Collaborative Research

Osama Zafar

*Department of Computer and Data Sciences
Case Western Reserve University
Cleveland, OH 44106, USA
oxz23@case.edu*

Rosemarie Santa González

*School of Computer Science
Georgia Institute of Technology
Atlanta, GA 30332, USA
rosemarie.santa@gatech.edu*

Mina Namazi*

*Center for Cybersecurity Research
Universitat Oberta de Catalunya
Barcelona, Spain
mnamaziesfanjani@uoc.edu*

Alfonso Morales

*Department of Planning and Landscape Architecture
University of Wisconsin–Madison
Madison, WI 53706, USA
morales1@wisc.edu*

Erman Ayday

*Department of Computer and Data Sciences
Case Western Reserve University
Cleveland, OH 44106, USA
exa208@case.edu*

Abstract—Data-driven agriculture, which integrates technology and data into agricultural practices, has the potential to improve crop yield, disease resilience, and long-term soil health. However, privacy concerns, such as adverse pricing, discrimination, and resource manipulation, deter farmers from sharing data, as it can be used against them. To address this barrier, we propose a privacy-preserving framework that enables secure data sharing and collaboration for research and development while mitigating privacy risks. The framework combines dimensionality reduction techniques and differential privacy by introducing noise to protect sensitive information. The proposed framework allows researchers to identify potential collaborators for a target farmer and to train personalized machine learning models either on the data of identified collaborators via federated learning or directly on aggregated, privacy-protected data. It also allows farmers to identify potential collaborators based on similarities. We have validated this on real-world datasets, demonstrating robust privacy protection against adversarial attacks and utility performance comparable to that of a centralized system. We demonstrate how this framework can facilitate collaboration among farmers and help researchers pursue broader research objectives. The adoption of the framework can empower researchers and policymakers to responsibly leverage agricultural data, paving the way for transformative advances in data-driven agriculture. By addressing critical privacy challenges, this work supports secure data integration, fostering innovation and sustainability in agricultural systems.

Index Terms—Privacy Enhancing Technologies, Digital Agriculture, Privacy Preserving Framework, Secure Collaborative Research

I. INTRODUCTION

AGRICULTURE has long been the backbone of human civilization, driving innovation to enhance food production and promote sustainability. Over the years, advances such

as fertilizers, pesticides, genetically modified seeds, and soil composition analysis have transformed cultivation and harvest practices [1], [2]. Today, this tradition of innovation is driven by a wealth of data collected from satellites, unmanned aerial vehicles, self-driving machines, and field sensors [3]. Although farmers occasionally share sales data for marketing purposes, sharing sensitive data on environmental and soil conditions presents significant privacy challenges [4]. The very data that empower precision agriculture to optimize yields and sustainability also carry risks of misuse, such as adverse pricing, price discrimination, disease interference, inflated insurance costs, and harmful competition [5], [6]. These concerns deter farmers from sharing their data, creating a critical barrier to advancing agricultural research. To unlock the full potential of data-driven agriculture, it is imperative to prioritize protecting farmers from these risks, ensuring that innovation does not come at the expense of their trust and livelihoods.

The existing data collection and collaboration frameworks in the domain, such as Farm2Facts [7], [8], Local Line [9], and Open Food Network [10], collect vast amounts of agricultural data but lack mechanisms for privacy-preserving collaboration. Publicly sharing sensitive agricultural data raises concerns about privacy and security, even for research purposes. As a result, farmers and researchers are unable to share data freely on such platforms, which impedes innovation. Several critical security threats in the digital agriculture domain pose significant risks to farmers' privacy and data security. These threats can be intentional attacks perpetrated by malicious third parties, such as attribute inference attacks and membership inference attacks, or they can be unintentional risks, including configuration errors and improper encryption, which may inadvertently expose sensitive information.

*This work was conducted while the author was with Case Western Reserve University.

This paper proposes a framework to address these privacy and security challenges in data-driven agriculture. The framework is designed to mitigate the risks of data misuse while fostering trust among farmers. It achieves this by employing techniques such as differential privacy and dimensionality reduction, which obscure sensitive data and prevent unauthorized reconstruction of the original datasets. Furthermore, the framework aims to reduce the risk of membership inference attacks by ensuring that even anonymized data remains untraceable to individuals. By addressing these specific threats, the framework creates a secure environment for data sharing and collaboration for the farmers, thereby promoting trust and enabling the agricultural research community to fully leverage the potential of data-driven agriculture.

The main goal of this work is to apply privacy-preserving machine learning tools, such as dimensionality reduction and differential privacy, to address a critical privacy challenge in digital agriculture: enabling collaborative research while safeguarding data privacy. Our research makes a significant contribution by developing a comprehensive privacy-preserving framework for agricultural collaboration. First, we introduce a novel combination of differentially private algorithms (such as Laplace noise addition) and dimensionality reduction techniques (such as Principal Component Analysis) to safeguard farmers' sensitive data while preserving its utility for collaborative research. Principal Component Analysis (PCA) reduces dimensionality from higher to lower dimensions, while Laplace Noise Addition ensures differential privacy. Although PCA does not inherently ensure privacy, its domain obfuscation of the original feature space complements privacy-preserving techniques, such as differential privacy, allowing for sensitive data analysis with reduced privacy risks. Integrating these two methods provides a strong security measure against threats such as membership inference attacks (see Section V for more details).

Building on this privacy foundation, we then apply clustering algorithms, such as K-Means, to identify and group farmers with similar characteristics, enabling the formation of networks for meaningful collaboration. By grouping farmers based on shared traits, such as crop types, soil composition, or farming practices, the framework facilitates knowledge exchange, best-practice sharing, and collective problem-solving. This allows farmers to connect with similar farmers based on the similarity of their farm profiles. This creates a collaborative environment that not only fosters peer learning but also empowers farmers to adopt innovative techniques, enhance productivity, and address common challenges through resource sharing and mutual support.

The proposed framework also achieves a robust balance between privacy preservation and utility, enabling effective training of ML models. It ensures that the transformed, privacy-protected data aggregated by the sandbox from farmers preserves the utility required for effective model training. Evaluation of the framework on real-world datasets such as the Wisconsin Farmers' Market and Crop Recommendation dataset [11] indicates that models trained using the proposed

architecture achieve accuracies comparable to those of models trained on centralized raw data (see details in Section III-D).

Our framework combines privacy preservation with a sandboxed collaboration environment, enabling researchers to derive insights without exposing raw farmer or market data and providing a novel solution for collaborative research in the agricultural domain, where such systems are absent despite widespread demand. We envision it to be deployed as a web-based sandbox environment housed on a cloud computing platform like AWS or Azure, featuring a user-friendly interface and an automated backend for data integration, data processing, model training, and sharing. The design ensures easy adoption by farmers and researchers without requiring technical expertise.

The remainder of this work is structured as follows. To lay the groundwork for our proposed solutions, we begin with a literature review of foundational concepts and related work in **Section II: Background and Related Work**. Next, **Section III: Proposed Solution** delves into the design and technical details of our framework. **Section IV: Security Analysis** then formalizes the privacy guarantees of the framework, presenting the security theorem and its proof. **Section V: Privacy and Utility Evaluation** evaluates privacy and utility on real-world datasets. **Section VI: Applications in Machine Learning** explores the application of our framework to real-world scenarios and demonstrates the practical utility of our approach. Finally, **Section VII: Conclusion** summarizes our contributions and discusses their potential impact.

II. BACKGROUND AND RELATED WORK

This section briefly presents literature relevant to our study, including data privacy challenges in digital agriculture, existing frameworks and mechanisms for secure data sharing, and data privacy techniques. Each provides insights into the current state of research and the gaps our work aims to address.

A. Agricultural Privacy Concerns

As the agricultural industry evolves, farmers are increasingly adopting technologies to manage soil, water, and harvest more efficiently. Since 2010, these innovations, collectively known as Climate-Smart Agriculture or precision agriculture, have transformed large-scale farming practices by enabling real-time monitoring and input optimization [12], [13]. These practices are critical for commodity farmers, improving resource use efficiency and sustainability by tailoring inputs to specific crop conditions. However, data-driven technologies also risk deepening existing inequalities in the agricultural sector [6], particularly among small and diversified farmers.

Small and medium-sized farmers face significant barriers to adopting data-driven technologies. First, these tools are predominantly designed for large-scale, monoculture farming systems, making them less applicable to diversified farms with varying crops and conditions [14], [15]. Second, while large-scale farmers often share data with industry partners to receive tailored recommendations, small-scale farmers are hesitant due to concerns over potential data misuse [6], [14], [16]. Since

2015, academic research has increasingly recognized these privacy concerns, proposing legislative protections and software solutions to alleviate these challenges [3], [17], [18]. Despite these efforts, as of August 2024, the agricultural sector lacks a dedicated federal regulatory body for data protection—unlike the healthcare sector (HIPAA) and the finance sector (Gramm-Leach-Bliley Act) [19]. While the U.S. Cybersecurity and Privacy Operations Center (CPOC) oversees some agricultural data concerns [20], it lacks the authority to enforce industry-wide protections [19].

Agricultural data storage involves multiple stakeholders, including Agricultural Technology Providers (ATPs), farmers, agronomists, and digital platforms. Emerging technologies, such as edge computing and blockchain, promise robust data security and traceability [21], [22]. However, small-scale farmers face substantial challenges in adopting these advanced systems, often bearing significant risks without receiving proportional benefits [3], [5]. Additionally, the complexity of lengthy, technical data-sharing agreements further discourages smallholder participation, exacerbating their concerns about data misuse and undermining trust in digital agriculture systems [23].

B. Data Sharing Frameworks for Digital Agriculture

The literature on data-sharing frameworks in agriculture has addressed privacy and security concerns. For instance, [24] proposes a set of principles and an access control mechanism within Data Sharing Agreements (DSAs) to regulate access to shared data and address privacy concerns in the agricultural sector. Similarly, [25] introduces a flexible, cloud-based, privacy-preserving data aggregation scheme that allows users to access aggregated plain-text data while safeguarding farmers' sensitive information. These frameworks offer scalable solutions for secure data sharing; however, their applicability to small and medium-sized farming operations remains limited.

Most studies focus heavily on securing smart devices from external attacks, often overlooking broader privacy frameworks tailored to diversified farming operations. For example, [26] proposes a multi-layered architecture that combines cyber-physical measures to deter cyberattacks on smart farming systems. [27] extends this work by introducing a privacy encoding framework for sensor-based systems, complemented by an intrusion detection system that maintains data integrity. Similarly, [28] developed a privacy-preserving framework for unmanned aerial vehicles (UAVs), emphasizing authentication mechanisms to prevent data poisoning attacks that could compromise agricultural decision-making processes.

Despite these advancements, there is a growing necessity for tailored security frameworks that address the unique challenges faced by small and medium-sized farms, particularly those with diverse farming practices [29]. To the best of our knowledge, few existing frameworks enable effective and privacy-preserving data collection from farmers, particularly for use in AI and machine learning applications. Our study addresses this critical gap by proposing and testing a privacy-

preserving framework that ensures the confidentiality of individual farmers' data while optimizing the utility of aggregated datasets. By balancing privacy and data usability, our framework empowers researchers to develop AI-driven solutions that are both ethical and impactful for agricultural research.

C. Relevant Data Privacy Techniques

In the broader context of data-driven approaches, protecting sensitive information is crucial across various domains, including agriculture. While some data privacy techniques have been widely applied in fields such as finance and healthcare, they also have potential to enhance data security in digital agricultural applications. This section examines several relevant key data privacy techniques, exploring both their theoretical foundations and their adaptability for use in agriculture. We will discuss PCA, which provides data abstraction by compressing the data into a lower dimension, thereby obfuscating the original feature space; and Local Differential Privacy (LDP), a method that ensures robust privacy through local data perturbation. Understanding the mathematical principles behind these techniques is essential for appreciating their role in safeguarding sensitive data, particularly as presented in the proposed framework.

Principal Component Analysis (PCA) [30] is a widely utilized technique for data analysis and the identification of patterns and relationships within data. It has a wide variety of applications, such as image compression [31]; facial recognition [32]; medical data correlation [33]; and Quantitative Finance [34]. PCA itself does not inherently guarantee privacy, as its primary function is to reduce data dimensionality by transforming it into orthogonal components. However, this transformation abstracts the data, making it harder to reverse engineer individual records and thereby providing a form of data obfuscation. Similar techniques have been used to identify population stratification in genomics research [35].

Local Differential Privacy (LDP) [36], [37] is a privacy-preserving technique that builds upon the principles of classic Differential Privacy (DP) [38]. LDP is designed to protect individual data and extends the concept of DP to a decentralized context. In LDP, each user perturbs their own data prior to sending it to a data collector. This local perturbation ensures that individual data privacy is protected, even when the data collector is not considered reliable. Unlike DP, which relies on a central, trusted party with access to the raw data, LDP offers privacy assurances directly at the data source. An algorithm A is considered to satisfy ϵ -LDP if, for any pair of individual private data points a_1 and a_2 , and for any possible output b :

$$\Pr[A(a_1) = b] \leq e^\epsilon \cdot \Pr[A(a_2) = b], \quad (1)$$

where ϵ represents the privacy parameter. ϵ -LDP is achieved by introducing an appropriate amount of noise to individual data points. The primary challenge is determining the magnitude of the noise to ensure compliance with LDP while preserving data utility. Various mechanisms have been developed in the DP field to address this problem, with the Laplacian mechanism [39] among the most prominent.

For any numerical function $f(x) : \mathbb{R} \rightarrow \mathbb{R}$ that satisfies ϵ -LDP if Laplacian noise is added as follows:

$$F(x) = f(x) + \text{Lap}\left(\frac{s}{\epsilon}\right) \quad (2)$$

where s represents the sensitivity of the function f and $\text{Lap}(k)$, where $k = \frac{s}{\epsilon}$, denotes sampling from a Laplace distribution with scale k . Sensitivity measures the maximum change that a single data point can cause to the output of f in the worst-case scenario. In our work, we calculate sensitivity to add Laplacian noise [40]. By definition, $s = \max_{x, x'} \|f(x) - f(x')\|_1$.

III. PROPOSED SOLUTION

A. System Model

Our proposed privacy-preserving scheme comprises two main actors: researchers R and participants P_i (e.g., farmers). Participants generate, own, and retain their data to derive insights and benefits from it. Researchers, on the other hand, need the data to conduct research analysis and to train machine learning models (see Table I). The relationship between researchers and participants is inherently symbiotic, as researchers need data to do their work, and ultimately, their work provides benefits to the farmers in the form of machine learning models, such as predictive models for disease detection or weather forecasting.

The entire process takes place within a secure framework that acts as a "sandbox environment." This framework handles crucial tasks in the middle, such as aggregating, clustering, and filtering the data. Together, these three parts—the researchers, the participants, and the secure framework—form the complete architecture of the system.

B. Threat Model

In the proposed framework, we assume the participants are trusted entities because they provide data to the researcher. They generate private agricultural data and perform local PCA transformations and noise addition (to achieve differential privacy for the shared data) before sharing the transformed data.

We assume the researcher to be an honest but curious party. They follow the protocol's specifications correctly. However, they might be curious to learn unauthorized information about participants' private data through protocol outputs, which include analyzing the transformed data matrices, clustering results, and machine learning models. The researcher has access to public agricultural datasets and can perform polynomial-time computations on all protocol outputs.

We also assume the framework to be honest but curious. Nonetheless, the protocol protects against data leakage by implementing differential privacy techniques. These techniques add noise to the data, obscuring individual farmers' sensitive information. The protocol also employs dimensionality-reduction methods, such as PCA, to further obfuscate the data structure, making it more challenging for attackers to extract meaningful insights from the interactions they observe.

TABLE I
FREQUENTLY USED NOTATIONS.

Notation	Description
R	Researcher
P_i	Participant i
D_s	Dataset used by the researcher to train the PCA model
M_s	Trained PCA model
O_i	Original PCA matrix of participant P_i
C_i	Noisy PCA matrix of participant P_i

We consider various adversarial attacks, including membership inference, attribute inference, and deanonymization. Since the identities of the participants who share the data are hidden, attribute inference attacks are only feasible if an attacker can infer the victim's membership in the research dataset through a membership inference attack. The proposed protocol is designed to be robust against membership inference attacks through several mechanisms. We evaluate the robustness of our framework against inference attacks by performing power analysis (see Section V). Although deanonymization attacks can be effective when auxiliary data is available, obscuring original feature space using dimensionality reduction and applying noise to shared datasets significantly reduces their success.

Each farmer independently performs data transformation and noise addition, ensuring that compromising one farmer's process cannot affect others. The framework infrastructure enforces a strict separation between different farmers' data streams, preventing unauthorized mixing or correlation of data from various sources. The framework infrastructure is assumed to maintain its integrity and not collude with any party to breach privacy guarantees.

C. Proposed Scheme

We propose a privacy-preserving mechanism that uses dimensionality reduction and differential privacy techniques to identify farmers with a specific set of attributes. Figures 1 and 2 illustrate the framework design and workflow. Figure 1 illustrates a high-level conceptual overview of the framework's architecture and workflow, free from technical distractions. It provides a high-level system figure, and it emphasizes the end-to-end process, focusing on roles, data flow, and anonymization at an abstract level to enhance clarity for the audience. Figure 2 presents a more detailed overview of the proposed framework, concentrating on the granular technical aspects of the core components, data flow, data processing, and infrastructure details essential for replicability and a deeper understanding. See Table I for the list of notations and their definitions.

The researcher R trains a global dimensionality reduction model, such as a PCA model, on a publicly available dataset D_s having desired farm features (e.g., nitrogen levels, annual rainfall, or pH). The global model is generated to ensure that participants can map their locally reduced data into a common space, enabling consistent, comparable results when the data is integrated or analyzed together. We use the Standard

Scaler function from scikit-learn [41] to preprocess the dataset D_s . This trained dimensionality-reduction model M_s is then shared with all participants via the sandbox F , along with the sensitivity parameter, which will be used later in the noise-addition step. Each P_i uses model M_s to transform its private data into the transformed matrix O_i . To guarantee data privacy, especially against membership inference attacks, the framework applies privacy enhancement techniques, such as locally adding Laplacian noise, to achieve ϵ -LDP. Noise is added to each component of a O_i matrix using 1, where sensitivity s is replaced with s_i , which denotes the sensitivity of component i in the O_i matrix. Finally, a differentially private version of their transformed data C_i is sent to sandbox F .

The sandbox environment receives differentially private data collected from multiple farmers, aggregates it, and then applies clustering algorithms to group farmers. Figure 2 presents the complete workflow of the framework, with a specific use case of connecting similar farmers based on farm attributes. Farmers interact with the sandbox by uploading their farm profile, which is then input to a clustering algorithm, such as K-Means [42], pre-trained on aggregated data. The clustering labels are utilized to group farmers and recommend connections among those with similar characteristics. Any P_i (farmer) can query the sandbox with their farm profile to identify potential collaborators using aggregated data. Farmers can communicate, share knowledge, and collaborate with recommended potential collaborators.

Similarly, the researcher R can utilize the aggregated data to query the sandbox with a target farm profile (e.g., similar to a farm P_i) to identify potential collaborators. This helps the researcher R to train various machine learning models on identified farmers' data. R identifies potential collaborators of a target P_i and uses Federated Learning techniques to train personalized machine learning models tailored to the target P_i profile, trained on data from identified potential collaborators. Furthermore, machine learning models can be trained directly on the aggregated data to make predictions. These models can predict various outcomes, such as the likelihood of farm success or failure, or analyze risks like disease outbreaks. It is important to emphasize that R does not have direct access to the private datasets. All analyses and predictions are conducted exclusively using differentially private outputs collected from farmers, ensuring a completely privacy-preserving approach.

D. Case Study

For feasibility testing and evaluation¹, the following case study explores the potential of the proposed privacy-preserving framework for connecting farmers based on attributes and evaluates it using two distinct datasets: a Wisconsin Farmer's Market dataset (Table II) and a crop recommendation dataset [11] (Table III). We consider the following use case from the domain of digital agriculture research: farmers seek to identify potential collaborators based on farm characteristics.

Farmer's Market dataset (Table II), which was gathered internally and has not been made public, includes detailed information about various aspects of farmer's market activities and vendors, and was explicitly collected for research purposes. Due to the private nature of the information contained in this dataset, it remains private. Meanwhile, the crop recommendation dataset (Table III) is publicly available and was collected from multiple farms, capturing soil composition and environmental factors [11]. The dataset is segmented into five distinct parts representing diverse markets, characterized by uneven sample distributions for each crop. Data is sourced from various farmers and shared with researchers. We envisage a scenario in which a researcher accesses this dataset using the aforementioned privacy-preserving techniques and employs a clustering algorithm, particularly K-Means. Figure 3 and Figure 4 illustrate K-Means clustering plots on the PCA-transformed farmer's market dataset in Table II and the crop recommendation dataset in Table III, respectively. To recommend potential collaborators, the farm profile is input into the K-Means model to ascertain the associated group label. Subsequently, the nearest neighbor algorithm [43] is utilized to identify the nearest neighbors of the farm profile within the same group label.

The entire process is repeated at the Farmer's Market dataset level to identify potential collaborators for personalized machine learning model training via federated learning (see Figure 1, which illustrates the end-to-end lifecycle of the framework, showcasing one of its use cases: how the researcher uses aggregated data to identify partners and train personalized models.). The crop recommendation dataset (Table III) is divided into a global training dataset and multiple farmers' market datasets to simulate experiments. For personalized model training, a participant P_i (a farmer's market in this case) initiates the training. The framework identifies three potential collaborators of the initiator P_i among the 5 who can contribute to training machine learning models in a federated learning setting, thereby enabling secure and collaborative research. There are two ways for a framework to identify potential collaborators: (i) based on the similarity of the participants' profiles (data) with each other, and (ii) based on the similarity of the distribution of participants' data. The framework uses the nearest-neighbor algorithm [43] to sort all participants in descending order of similarity and selects the first three. The trained model is personalized for the initiator participant. Figure 5 and Figure 6 illustrate the accuracy of the federated learning model vs. the epsilon privacy budget for a neural network trained on three identified collaborators of the initiator P_i . In Figure 5, the collaborators are selected based on the similarity of their participant profiles, while in Figure 6, the collaborators are chosen based on the similarity of the distribution of their data.

IV. SECURITY ANALYSES

The security of the proposed framework reduces to the security of the underlying ϵ -LDP mechanism. In the following, we provide the formal security theorem and its proof.

¹All implementation codes are available on GitHub and will be provided upon request.

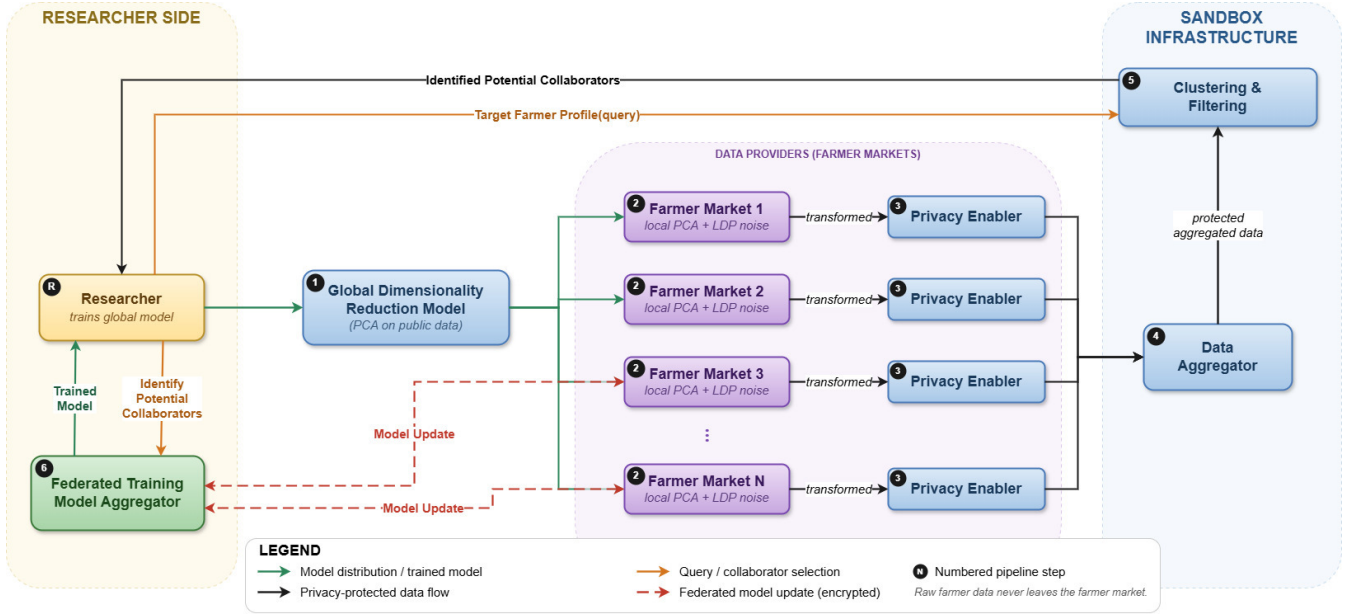


Fig. 1. A high-level data flow: the researcher sends a globally trained dimensionality reduction model to N farmer markets, who transform and anonymize their data before sending it to a data aggregator for sharing on the sandbox platform. The researcher then identifies potential collaborators to train a personalized model.

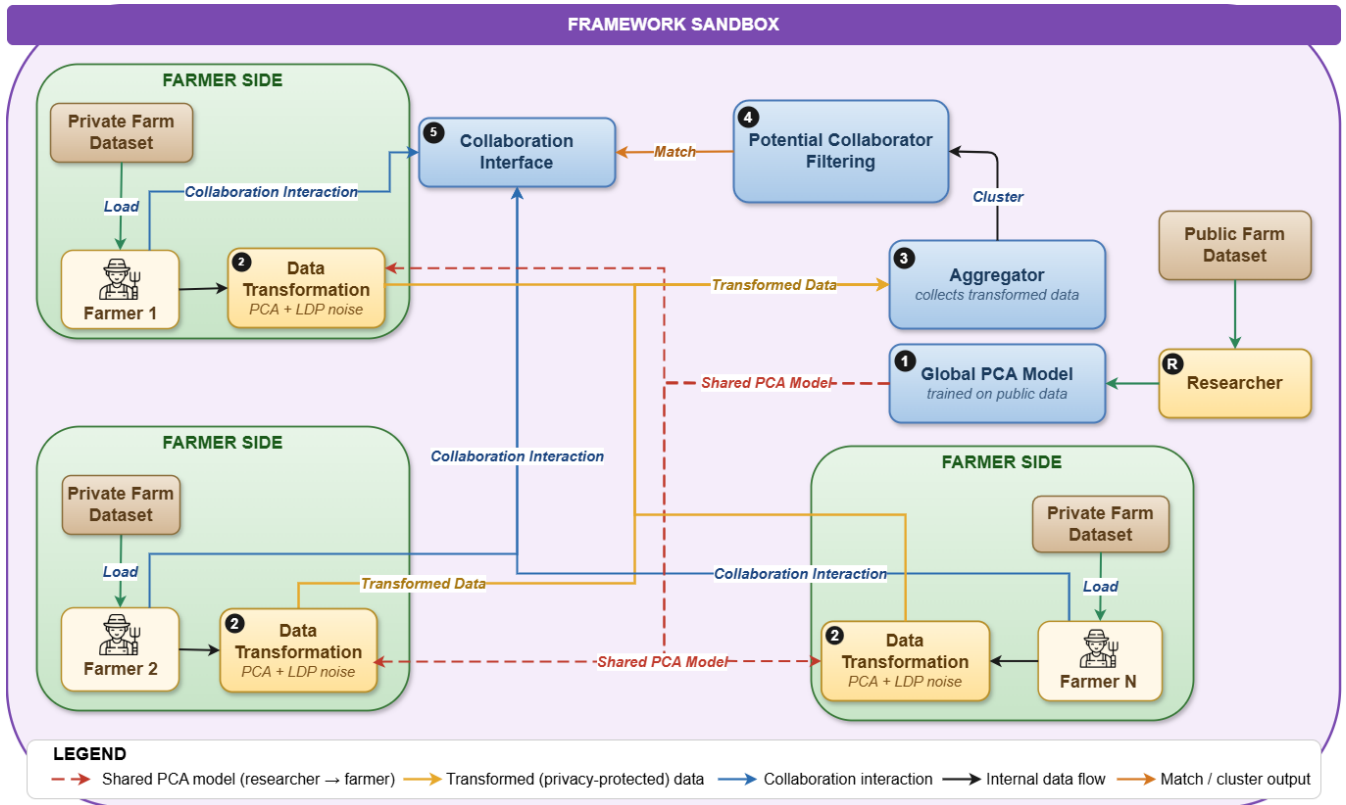


Fig. 2. Detailed architecture of the proposed framework.

Theorem 1. Let M_s be the randomization mechanism (e.g., Laplacian mechanism) used in the reduction model. The privacy-preserving scheme for the collaborative agriculture research sharing framework is secure and protects the privacy

TABLE II
FARMERS MARKET DATASET.

Miles from Market		Vendor Type									Sales	#Visitors
ID	Primary Location	F&V	M&S	D	Eggs	P&F	N&L	VA	PF	CAS		
6815	42.22	0	0	1	0	0	0	1	1	0	414.0	70
5991	19.62	1	1	0	0	0	0	0	0	0	157.0	404
5663	23.54	0	0	0	0	0	0	0	0	1	670.0	6
5950	70.52	0	0	0	0	0	0	1	0	0	70.0	53
5686	0.68	0	0	0	0	0	0	0	0	0	285.0	166

F&V: Fruits and Vegetables; M&S Meat and Seafood; D:Dairy; P&F: Plants and Flowers; N&L: Nuts and Legumes;
VA: Value-added; PF: Prepared Food; CAS: Crafts/Art/Services.

TABLE III
CROP RECOMMENDATION DATASET.

N	P	K	Temp	H	pH	R	Label
90	42	43	20.879	82.002	6.502	202.935	rice
61	44	17	26.100	71.574	6.931	102.266	maize
40	72	77	17.024	16.988	7.485	88.551	chickpea
13	60	25	17.1636	20.595	5.685	128.256	kidney beans
90	46	42	23.978	81.450	7.502	250.082	rice

N: Nitrogen; P: Phosphorous; K:Potassium; pH: pH value; R: rainfall in mm;
Temp: temperature in Celsius; H: relative humidity in %.

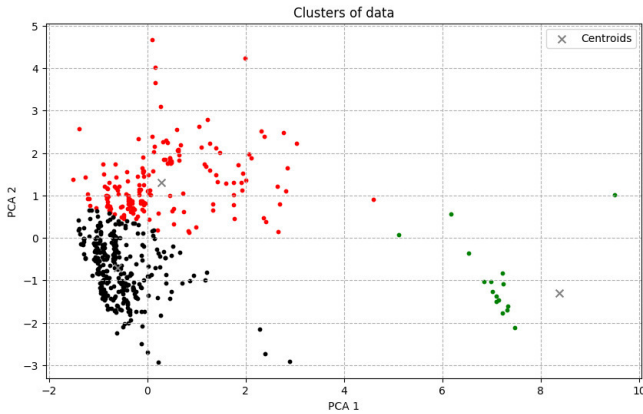


Fig. 3. PCA transformed farmer's market dataset clustering.



Fig. 4. PCA transformed crop recommendation dataset clustering.

of the farmers against PPT semi-honest adversaries if the PCA transformation and Laplacian noise satisfies ϵ -Local Differential Privacy.

We prove theorem 1 by introducing two worlds. There is an ideal world the adversary's view $\text{View}_A^{\text{Ideal}}$ is the outputs of the interactions with a simulator who has no access to the datasets. Moreover, in the real world, the adversary's view $\text{View}_A^{\text{Real}}$ is the actual output of the framework using the reduction model and Laplacian noise mechanism. The ϵ -LDP guarantee

ensures that the outputs of the mechanism for neighboring datasets are statistically close (up to a factor of e^ϵ), meaning that the adversary's view in the ideal world is computationally indistinguishable from the real world view.

$$\text{View}_A^{\text{Real}} \stackrel{c}{\approx} \text{View}_A^{\text{Ideal}}$$

Where $\stackrel{c}{\approx}$ indicates that no PPT adversary A can distinguish between the two views with non-negligible advantage.

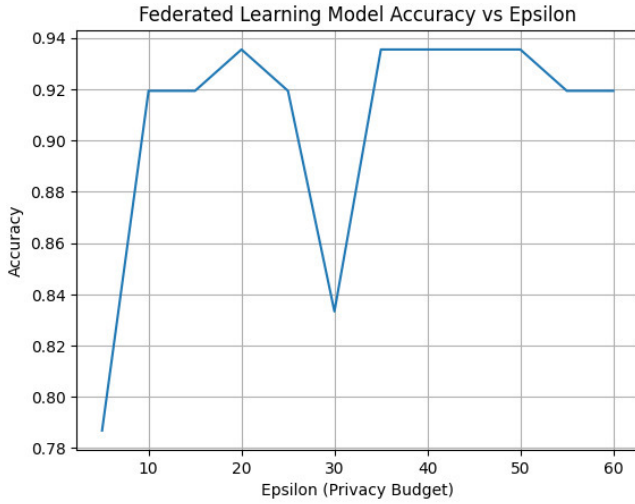


Fig. 5. Federated learning accuracy vs. epsilon (based on profile similarity).

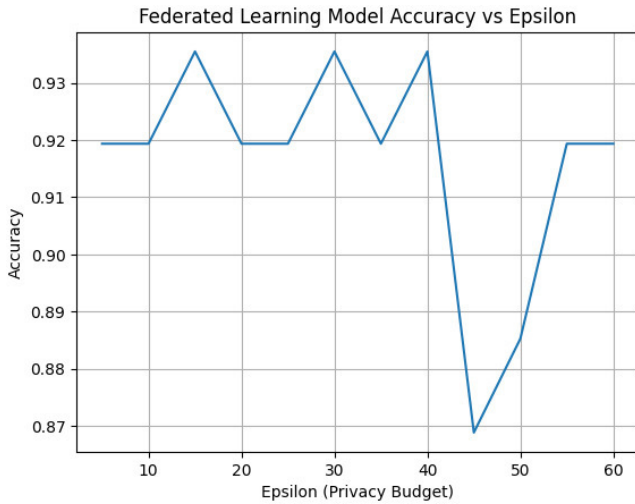


Fig. 6. Federated learning accuracy vs. epsilon (based on data distribution similarity)

Proof. In the ideal world, the simulator S accesses the privacy parameter ϵ and the adversary's queries and cannot access the actual datasets D_s^0 or D_s^1 . For each query made by the adversary, S generates noisy outputs using an ideal ϵ -LDP mechanism. The outputs are indistinguishable from those in the real world because of the ϵ -LDP guarantee.

The real world interacts with the adversary A by answering their queries and providing noisy outputs according to the actual interactions of the privacy-preserving framework.

If A has a non-negligible advantage in distinguishing between the two worlds, we can construct a distinguisher D that breaks the ϵ -LDP guarantee of the mechanism M_s .

The distinguisher D receives the output C_i generated from M_s for one of two neighboring datasets D_s^0 or D_s^1 and simulates the real-world mechanism using C_i as the output

of M_s . Then, it interacts with the adversary A to send A 's queries to M_s , generating noisy outputs as needed. D observes A 's decision and uses it to distinguish which dataset C_i was generated from. If A can distinguish between the real and ideal worlds with a non-negligible advantage, then D can distinguish between the outputs of M_s applied to D_s^0 and D_s^1 , contradicting the ϵ -LDP guarantee.

By contradiction, A cannot distinguish between the real and ideal worlds with non-negligible advantages. Therefore, the outputs of the real-world mechanism are computationally indistinguishable from those of the ideal-world mechanism, and the privacy-preserving framework is secure under the ϵ -LDP assumption, which concludes the proof. $\square$

V. PRIVACY AND UTILITY EVALUATION

Researchers can use a privacy-preserving framework to identify a participant's nearest neighbors, or potential collaborators, to train models on their raw data using Federated Learning. To evaluate the effectiveness of the model (trained on data of identified potential collaborators of a particular participant), its performance is compared to a model trained on all available farmer datasets. We consider the Wisconsin Farmer's Market dataset (See Table II) from our case study section. To simulate two local farmers' market datasets P_i s and the dataset used by the researcher to train the PCA model D_s , we divide our dataset into three parts. Each P_i converts its data to C_i using the shared PCA model M_s , which is accessible to the sandbox.

To evaluate the utility of our process, we compare the accuracy of a classifier trained on raw data aggregated from all farmers in a centralized manner vs. a classifier trained on data aggregated through the proposed architecture. Table IV illustrates the accuracies of various models trained under both settings. The models trained using the proposed architecture exhibit performance comparable to that of the traditionally trained models on centralized data, with an average accuracy loss of 5.33%. This difference represents a reasonable trade-off, given the substantial privacy benefits achieved.

To evaluate the effect of noise addition on the utility of the data, we compared model performance on aggregated data with and without noise across varying values of epsilon (ϵ). The parameter ϵ quantifies privacy protection, where smaller values of ϵ correspond to stronger privacy guarantees and higher noise, leading to a trade-off with utility. Figures 7, 8, and 9 illustrate the power (fraction of correctly identified noisy samples) and accuracy for a range of ϵ values for the Logistic Regression, Naive Bayes, and Support Vector Machine (SVM) algorithms, respectively on the crop recommendation dataset.

To evaluate the privacy performance of our algorithm, we compute the power metric, which measures the fraction of noisy samples correctly identified. To achieve this, we calculate the minimum pairwise distances between the samples of the shared dataset and the control group as the control distances. Similarly, we compute the minimum pairwise distance between the shared dataset and each case group as

TABLE IV
ACCURACIES OF MODELS TRAINED ON CENTRALIZED DATA VS PROTECTED AGGREGATED DATA AT IDEAL EPSILON VALUES.

Model Name	Accuracy on Centralized Data	Accuracy on Aggregated Data
Logistic Regression	99.6	92.1
Naive Bayes	98.3	93.8
Support Vector Machine	99.5	95.5

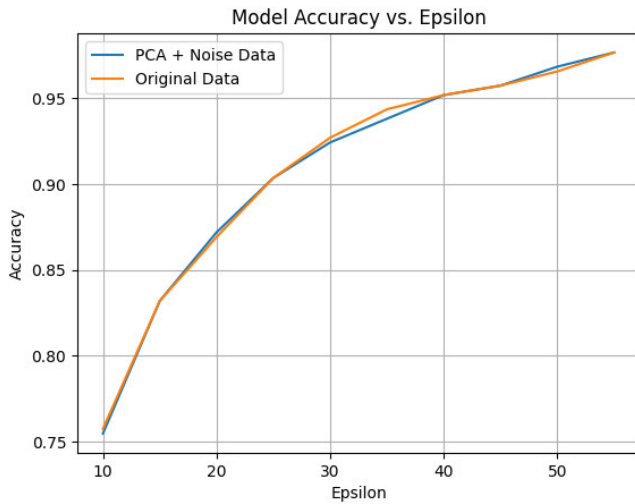


Fig. 7. Model accuracy on noisy and original data (Logistic Regression) vs epsilon on crop recommendation dataset.

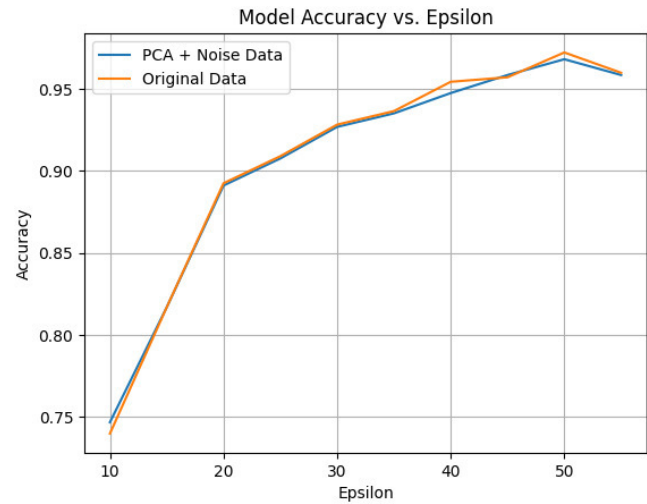


Fig. 8. Model accuracy on noisy and original data (Naive Bayes) vs epsilon on crop recommendation dataset.

the case distance. Here, the control group is sampled from the sandbox global dataset, and the case group is sampled from the participant's shared dataset. Then, we determine the threshold, defined as the distance at the 95th percentile (when the False Positive Rate is 5%) of the sorted control distances. This threshold is subsequently used to evaluate the fraction of case distances that fall below it, thereby determining the power. The metric is evaluated using specific thresholds for the False Positive Rate (FPR) and the epsilon (ϵ). ϵ is a privacy parameter for privacy loss due to a differential alteration in the dataset and is allocated across columns according to their sensitivity. For our analysis, we set the FPR to 0.05 (5%) and incrementally vary ϵ , starting from 10, to observe its impact on the power metric. This approach helps us select the correct value of ϵ with acceptable robustness, as measured by the power metric.

For the utility metric, we compare the noise-free cluster labels with the labels predicted by a binary classifier for noisy samples, assessing the accuracy of that prediction. We have performed the aforementioned utility test using three different algorithms (Logistic Regression, Naive Bayes, and Support Vector Machine) to demonstrate the results across classifiers. Figures 13, 14, and 15 illustrate the power and accuracy for a range of ϵ values for the Logistic Regression, Naive Bayes, and

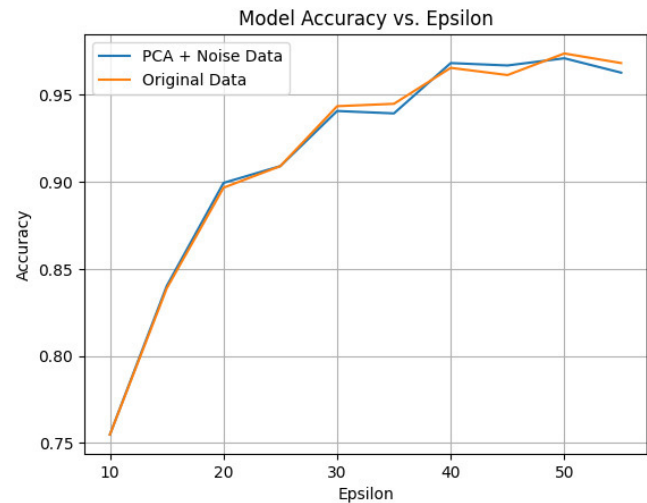


Fig. 9. Model accuracy on noisy and original data (SVM) vs epsilon on crop recommendation dataset.

SVM algorithms, respectively, on the farmer's market dataset. Similarly, Figures 10, 11, and 12 illustrate the power and accuracy for a range of ϵ values for the Logistic Regression, Naive Bayes, and SVM (Support Vector Machine) algorithms,

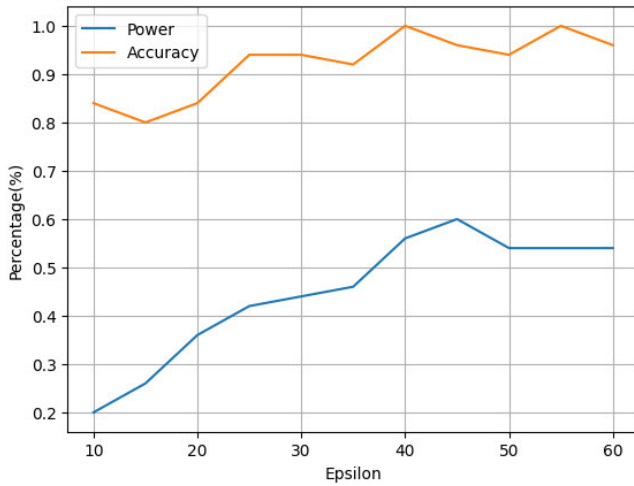


Fig. 10. Power and accuracy (Logistic Regression) vs epsilon on crop recommendation dataset.

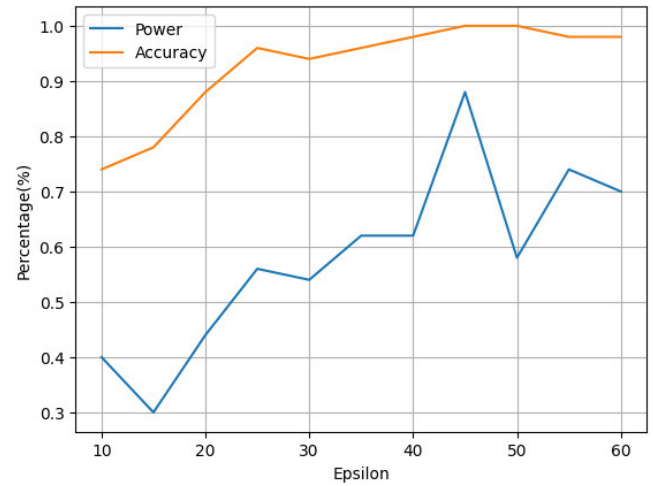


Fig. 12. Power and accuracy (SVM) vs epsilon on crop recommendation dataset.

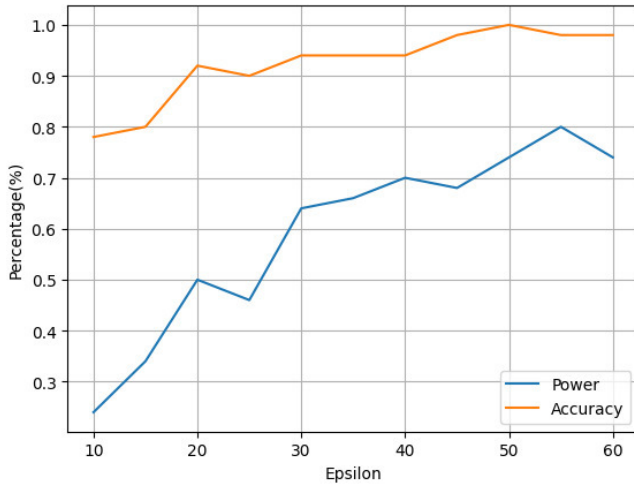


Fig. 11. Power and accuracy (Naive Bayes) vs epsilon on crop recommendation dataset.

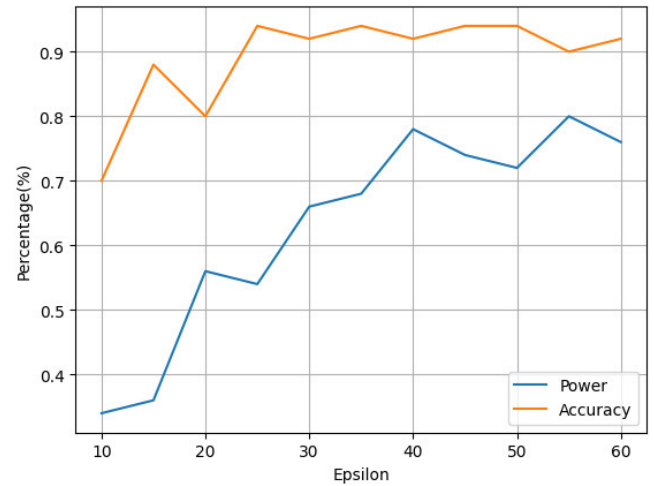


Fig. 13. Power and accuracy (Logistic Regression) vs epsilon on farmer's market dataset.

respectively on the crop recommendation dataset. The results reflect the high robustness of privacy with comparable utility accuracy. For each model, the optimal ϵ is determined by a trade-off between privacy (measured by the power metric) and utility (measured by accuracy). Specifically, the optimal ϵ values correspond to those that achieve both low power (indicating strong privacy protection) and high accuracy. Therefore, optimal epsilon values are 25, 35, and 35 for the Logistic Regression, Naive Bayes, and Support Vector Machine models, respectively.

Overall, the framework achieves comparable model accuracies to centralized data training, with minimal utility loss across diverse classifiers (see Table IV). This demonstrates that privacy-preserving techniques can be effectively integrated into machine learning models without compromising performance. Operating under realistic assumptions of non-colluding

participants and honest-but-curious researchers, a common model in privacy-preserving machine learning, the framework significantly mitigates the effectiveness of adversarial attacks like membership inference attacks, providing a robust defense against privacy breaches. The use of the PCA algorithm transforms data into principal components, thereby obfuscating the original feature space and protecting the original data from reconstruction attacks.

Additionally, the analysis of noise addition and the power metric further confirms that optimal ϵ values yield comparable accuracies across various machine learning models, demonstrating the framework's robustness against adversarial attacks and underscoring the need for real-world collaborative machine learning applications.

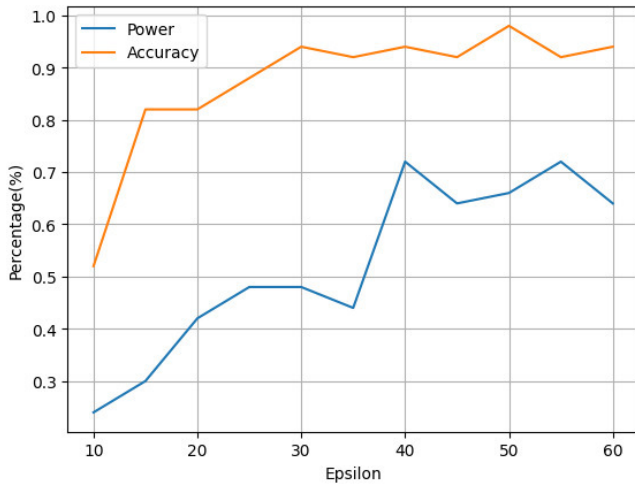


Fig. 14. Power and accuracy (Naive Bayes) vs epsilon on farmer's market dataset.

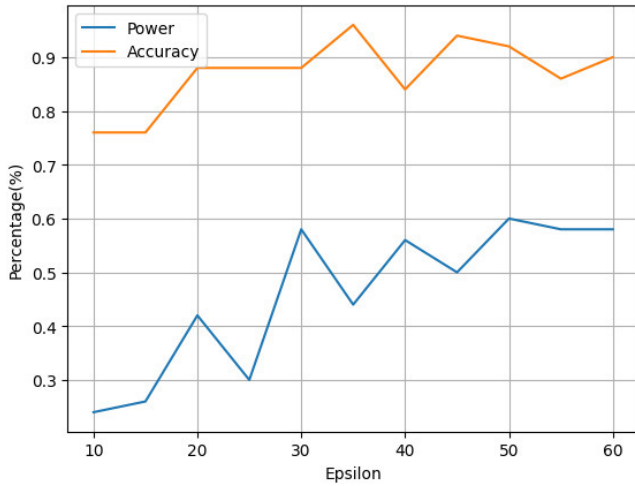


Fig. 15. Power and accuracy (SVM) vs epsilon on farmer's market dataset.

VI. APPLICATIONS IN MACHINE LEARNING

The proposed framework enables researchers to aggregate private data from multiple farmers, facilitating the development of a diverse array of machine learning models that significantly improve the breadth and depth of agricultural research. Building upon this foundation, various detailed examples from the case study dataset illustrate how researchers can create advanced models. For example, they can predict the most suitable crop type for a specific farm by analyzing factors such as soil composition, humidity levels, temperature, and other environmental variables. Similarly, they can leverage farmers' market data to develop models that forecast farm success based on parameters like farm size, proximity to markets and visitors, and historical sales data. Additional models can be designed to assess food accessibility by considering factors such as distance from markets, visitor traffic logs, and regional

public insecurity levels. Further applications include modeling and predicting soil conditions over time to recommend effective management practices and estimate regional crop yields; integrating weather data with soil quality, moisture levels, and water needs to refine irrigation strategies; and evaluating the potential impacts of climate change on crop productivity and soil health. These insights support adaptive agricultural planning and sustainability initiatives, ultimately enhancing resilience and efficiency in farming practices.

VII. CONCLUSION

Privacy concerns, especially among farmers, hinder the data sharing for precision agriculture, despite its potential for research and innovation. A robust, privacy-preserving data-sharing solution is, therefore, essential to balance the need for innovation with the protection of sensitive information. This study presents a novel framework that effectively addresses this challenge by employing techniques such as dimensionality reduction and differential privacy. The framework safeguards individual privacy while enabling researchers to leverage aggregated data for machine learning applications without compromising data utility.

Our contributions advance the field by introducing a privacy-preserving framework that provides both security and usability. The framework also facilitates the identification and grouping of farmers with shared characteristics through clustering algorithms, fostering collaboration and collective problem-solving among farmers with similar needs. By addressing these critical gaps, this work provides a scalable and ethical solution for advancing AI-driven research while empowering the agricultural community.

Despite its strengths, the framework has limitations. For instance, the computational requirements of differential privacy techniques may pose challenges for widespread adoption among resource-constrained farming operations. Additionally, the framework's reliance on consistent and high-quality input data may limit its effectiveness in some regions. Future work will focus on refining the framework to enhance its scalability and adaptability to diverse agricultural systems. In addition, further research will explore the integration of decentralized data-sharing mechanisms and real-time data processing to improve trust and responsiveness. By addressing these challenges, this work represents an important step toward enabling privacy-preserving and equitable digital agriculture, laying a foundation for sustainable and ethical farming practices that benefit both researchers and farmers.

VIII. ACKNOWLEDGMENTS

This research was supported in part by the NSF (awards OAC-2112606, 2427505) and the USDA (grant NR233A750004G019). It was also supported by the Spanish Ministry of Science and Innovation (AEI/ERDF) under grants SECURING (PID2021-125962OB-C31) and SAFE (PID2024-156914OB-C4), and by INCIBE (EU-NextGenerationEU/PRTR) through the ARTEMISA Chair (C057/23) and DANGER Project (C062/23).

REFERENCES

- [1] J. Pretty, "Agricultural sustainability: concepts, principles and evidence," *Philosophical Transactions of the Royal Society B: Biological Sciences*, vol. 363, no. 1491, pp. 447–465, 2008. [Online]. Available: <https://doi.org/10.1098/rstb.2007.2163>
- [2] D. Tilman, K. G. Cassman, P. A. Matson, R. Naylor, and S. Polasky, "Agricultural sustainability and intensive production practices," *Nature*, vol. 418, no. 6898, pp. 671–677, Aug 2002. [Online]. Available: <https://doi.org/10.1038/nature01014>
- [3] S. Wolfert, L. Ge, C. Verdouw, and M.-J. Bogaardt, "Big data in smart farming – a review," *Agricultural Systems*, vol. 153, pp. 69–80, 2017. [Online]. Available: <https://doi.org/10.1016/j.agsy.2017.01.023>
- [4] C. DeMets and A. Morales, "Marketplace measurement: farmers, farmers markets and ecosystem services," *International Journal of Sociology and Social Policy*, vol. 42, 03 2022.
- [5] L. Wiseman, J. Sanderson, A. Zhang, and E. Jakku, "Farmers and their data: An examination of farmers' reluctance to share their data through the lens of the laws impacting smart farming," *NJAS-Wageningen Journal of Life Sciences*, vol. 90, p. 100301, 2019. [Online]. Available: <https://doi.org/10.1016/j.njas.2019.04.007>
- [6] M. Taylor, "Climate-smart agriculture: what is it good for?" *The Journal of Peasant Studies*, vol. 45, no. 1, pp. 89–107, 2018.
- [7] "Farm2facts." [Online]. Available: <https://farm2facts.org/>
- [8] E. Ledesma, A. He, P. Warsaw, L. Suerth, A. Morales, L. Rosenblum, and B. Wiedenfeld, "Citizen scientist: Farm 2 facts supporting farmers markets," *Sustainability*, vol. 13, no. 11, 2021. [Online]. Available: <https://www.mdpi.com/2071-1050/13/11/6162>
- [9] "Local line: The farm to fork commerce platform." [Online]. Available: <https://www.localline.co/>
- [10] "The open food network." [Online]. Available: <https://openfoodnetwork.org/>
- [11] "Crop recommendation dataset," accessed: 2024. [Online]. Available: <https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset>
- [12] FAO, "Fao.org," 2024. [Online]. Available: <https://www.fao.org/climate-smart-agriculture/overview/faqs/history/en/>
- [13] J. Lowenberg-DeBoer and B. Erickson, "Setting the record straight on precision agriculture adoption," *Agronomy Journal*, vol. 111, no. 4, pp. 1552–1569, 2019. [Online]. Available: <https://doi.org/10.2134/agronj2018.12.0779>
- [14] R. B. Wakweya, "Challenges and prospects of adopting climate-smart agricultural practices and technologies: Implications for food security," *Journal of Agriculture and Food Research*, vol. 14, p. 100698, 2023. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2666154323002053>
- [15] L. Klerkx, E. Jakku, and P. Labarthe, "A review of social science on digital agriculture, smart farming and agriculture 4.0: New contributions and a future research agenda," *NJAS-Wageningen journal of life sciences*, vol. 90, p. 100315, 2019.
- [16] O. Westermann, W. Förch, P. Thornton, J. Körner, L. Cramer, and B. Campbell, "Scaling up agricultural interventions: Case studies of climate-smart agriculture," *Agricultural Systems*, vol. 165, pp. 283–293, 2018. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0308521X1830043X>
- [17] S. Linsner, F. Kuntke, E. Steinbrink, J. Franken, and C. Reuter, "The role of privacy in digitalization—analyzing perspectives of german farmers," *Proceedings on Privacy Enhancing Technologies*, 2021.
- [18] J. Kaur, S. M. Hazrati Fard, M. Amiri-Zarandi, and R. Dara, "Protecting farmers' data privacy and confidentiality: Recommendations and considerations," *Frontiers in Sustainable Food Systems*, vol. 6, 2022. [Online]. Available: <https://www.frontiersin.org/journals/sustainable-food-systems/articles/10.3389/fsufs.2022.903230>
- [19] J. L. Ferris, "Data privacy and protection in the agriculture industry: Is federal regulation necessary?" *Minn. J.L. Sci. & Tech.*, vol. 18, p. 309, 2017.
- [20] 2024. [Online]. Available: <https://www.usda.gov/ocio/centers/cpoc>
- [21] M. Amiri-Zarandi, R. A. Dara, E. Duncan, and E. D. Fraser, "Big data privacy in smart farming: a review," *Sustainability*, vol. 14, no. 15, p. 9120, 2022. [Online]. Available: <https://doi.org/10.3390/su14159120>
- [22] A. Kamilaris, A. Fonts, and F. X. Prenafeta-Boldó, "The rise of blockchain technology in agriculture and food supply chains," *Trends in food science & technology*, vol. 91, pp. 640–652, 2019.
- [23] K. Kosior, "Digital transformation in the agri-food sector—opportunities and challenges," *Roczniki (Annals)*, vol. 2018, no. 2, 2018.
- [24] K. Spanaki, E. Karafili, and S. Despoudi, "Ai applications of data sharing in agriculture 4.0: A framework for role-based data access control," *International Journal of Information Management*, vol. 59, p. 102350, 2021. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0268401221000438>
- [25] J. Song, Q. Zhong, W. Wang, C. Su, Z. Tan, and Y. Liu, "Fpdp: Flexible privacy-preserving data publishing scheme for smart agriculture," *IEEE Sensors Journal*, vol. 21, no. 16, pp. 17430–17438, 2021. [Online]. Available: <http://doi.org/10.1109/JSEN.2020.3017695>
- [26] M. Gupta, M. Abdelsalam, S. Khorsandroo, and S. Mittal, "Security and privacy in smart farming: Challenges and opportunities," *IEEE Access*, vol. 8, pp. 34564–34584, 2020. [Online]. Available: <http://doi.org/10.1109/ACCESS.2020.2975142>
- [27] P. Kumar, G. P. Gupta, and R. Tripathi, "Pefl: Deep privacy-encoding-based federated learning framework for smart agriculture," *IEEE Micro*, vol. 42, no. 1, pp. 33–40, 2022.
- [28] R. Kumar, P. Kumar, R. Tripathi, G. P. Gupta, T. R. Gadekallu, and G. Srivastava, "Sp2f: A secured privacy-preserving framework for smart agricultural unmanned aerial vehicles," *Computer Networks*, vol. 187, p. 107819, 2021. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1389128621000086>
- [29] P. A. Ongadi, "A comprehensive examination of security and privacy in precision agriculture technologies," *GSC Advanced Research and Reviews*, vol. 18, no. 1, pp. 336–363, 2024.
- [30] K. P. F.R.S., "Liili. on lines and planes of closest fit to systems of points in space," *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, vol. 2, no. 11, pp. 559–572, 1901.
- [31] V. S. V. Gaidhane and M. Kumar, "Image compression using pca and improved technique with mlp neural network," *IEEE International Conference on Advances in Recent Technologies in Communication and Computing*, pp. 106–110, 2010.
- [32] R. Gottumukkall and V. Asari, "An improved face recognition technique based on modular pca approach," *Pattern Recognition Letters*, vol. 25, pp. 429–436, 03 2004.
- [33] N. Qureshi, V. Suthar, H. Magsi, M. Sheikh, M. Pathan, and B. Qureshi, "Application of principal component analysis (pca) to medical data," *Indian Journal of Science and Technology*, vol. 10, pp. 1–9, 02 2017.
- [34] H. Yu, R. Chen, and G. Zhang, "A svm stock selection model within pca," *Procedia Computer Science*, vol. 31, p. 406–412, 12 2014.
- [35] L. Dervishi, W. Li, A. Halimi, X. Jiang, J. Vaidya, and E. Ayday, "Privacy preserving identification of population stratification for collaborative genomic research," *Bioinformatics*, vol. 39, no. Supplement_1, pp. i168–i176, 06 2023. [Online]. Available: <https://doi.org/10.1093/bioinformatics/btad274>
- [36] J. Duchi, M. Jordan, and M. Wainwright, "Local privacy and statistical minimax rates," *Foundations of Computer Science, 1975., 16th Annual Symposium on*, 02 2013.
- [37] P. Kairouz, S. Oh, and P. Viswanath, "Extremal mechanisms for local differential privacy," vol. 17, 04 2016.
- [38] C. Dwork, "Differential privacy: A survey of results," vol. 4978, 04 2008, pp. 1–19.
- [39] C. Dwork, F. McSherry, K. Nissim, and A. Smith, "Calibrating noise to sensitivity in private data analysis," vol. 3876, 01 2006, pp. 265–284.
- [40] C. Dwork and A. Roth, "The algorithmic foundations of differential privacy," *Foundations and Trends in Theoretical Computer Science*, vol. 9, 01 2013.
- [41] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and Édouard Duchesnay, "Scikit-learn: Machine learning in python," *Journal of Machine Learning Research*, vol. 12, no. 85, pp. 2825–2830, 2011. [Online]. Available: <http://jmlr.org/papers/v12/pedregosa11a.html>
- [42] J. A. Hartigan and M. A. Wong, "Algorithm as 136: A k-means clustering algorithm," *Journal of the Royal Statistical Society. Series C (Applied Statistics)*, vol. 28, no. 1, pp. 100–108, 1979. [Online]. Available: <http://www.jstor.org/stable/2346830>
- [43] T. Cover and P. Hart, "Nearest neighbor pattern classification," *IEEE Transactions on Information Theory*, vol. 13, no. 1, pp. 21–27, 1967.